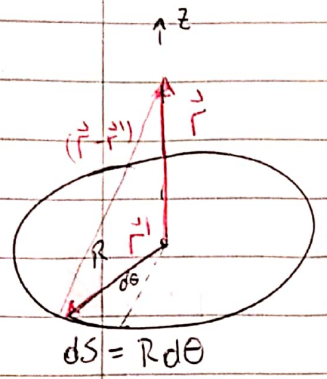
GUÍA 3

(1) Primero, se determina $\vec{E}$ y luego $\vec{F} = q\vec{E}$, considerando en éste último el signo de q. La densidad de carga es:



$$dq = \lambda R\, d\theta$$

$$Q = \int_0^{2\pi} \lambda R\, d\theta$$

$$4{,}5 \cdot 10^{-4} = \lambda \cdot 0{,}07 \cdot 2\pi$$

$$\lambda = 1{,}023 \cdot 10^{-3}\ [C/m]$$

$$= 1{,}023\ [mC/m].$$

$$\lambda = \frac{dq}{dS}$$

Luego, el campo eléctrico es:

$$d\vec{E} = K \frac{dq}{\|\vec{r}-\vec{r}'\|^3} (\vec{r}-\vec{r}')$$

$$\vec{r} = z\hat{k}$$

$$\vec{r}' = R\cos\theta\,\hat{i} + R\,\mathrm{sen}\,\theta\,\hat{j}$$

$$\vec{r}-\vec{r}' = -R[\cos\theta\,\hat{i} + \mathrm{sen}\,\theta\,\hat{j}] + z\hat{k}$$

$$\|\vec{r}-\vec{r}'\|^3 = (R^2 + z^2)^{3/2}$$

Escaneado con CamScanner

$$d\vec{E} = K \frac{\lambda R d\theta}{(R^2+z^2)^{3/2}} \left[-R(\cos\theta\,\hat{\imath} + \operatorname{sen}\theta\,\hat{\jmath}) + z\,\hat{k} \right]$$

$$\vec{E} = \frac{K\lambda R}{(R^2+z^2)^{3/2}} \left\{ -R \left[\int_0^{2\pi} \cancel{\cos\theta\, d\theta}^{\,0}\, \hat{\imath} + \int_0^{2\pi} \cancel{\operatorname{sen}\theta\, d\theta}^{\,0}\, \hat{\jmath} \right] + z \int_0^{2\pi} d\theta\, \hat{k} \right\}$$

$$= K \frac{\lambda R z}{(R^2+z^2)^{3/2}}\, 2\pi\, \hat{k}$$

$\Rightarrow$ En $z = 0{,}15$ m,

$$\vec{E} = 9 \cdot 10^9 \cdot \frac{1{,}023 \cdot 10^{-3} \cdot 0{,}07 \cdot 0{,}15 \cdot 2\pi}{(0{,}07^2 + 0{,}15^2)^{3/2}}\, \hat{k}$$

$$= 1{,}3392 \cdot 10^8\, \hat{k} \ [N/C]$$

$$\Rightarrow \vec{F} = -5 \cdot 10^{-6} \cdot 1{,}3392 \cdot 10^8\, \hat{k}$$

$$= -669{,}6\, \hat{k} \ [N].$$

Escaneado con CamScanner